Attention-Based Contrastive Learning for Few-Shot Remote Sensing Image Classification

Yulong Xu, Hanbo Bi, Hongfeng Yu, Wanxuan Lu, Peifeng Li, Xinming Li, and Xian Sun, *Senior Member, IEEE*

***Abstract*— Few-shot remote sensing image classification entails identifying images using a limited set of labeled data within remote sensing scenes, holding significant theoretical and practical implications. However, owing to the intricacy and variety of remote sensing images, traditional classification methods usually struggle to extract effective features and learn robust classifiers. To address this issue, an end-to-end metric learning framework named attention-based contrastive learning network (ACL-Net) is introduced in this article. Specifically, the attention-based feature optimization (ABFO) module is employed to align and enhance target image features, highlighting the target region and strengthening the network's feature extraction capability. In addition, the dictionary-based contrastive loss (DBCL) module is assigned to optimize image feature vectors, improving category distinguishability and consequently enhancing classification accuracy. The experimental results on five publicly available few-shot remote sensing classification datasets demonstrate the high competitiveness of our proposed method. Furthermore, it illustrates superior classification accuracy compared to other pertinent few-shot learning algorithms in the five-way one-shot scenario.**

***Index Terms*— Contrastive learning, few-shot learning, image scene classification, remote sensing.**

I. Introduction

REMOTE sensing image classification plays a pivotal role in the remote sensing image processing workflow. It assigns labels to regions of interest or those with significant representation in remote sensing images, simplifying data organization and analysis processes for human operators. As remote sensing technology continues to advance, remote sensing images are progressively becoming more intricate and multilayered, leading to a wealth of detailed information within the images [1], [2], [3], [4]. However, the presence of noise, occlusion, and background interference factors in remote sensing images significantly impede the model's ability to distinguish categories [5], [6], [7]. Moreover, in some remote sensing scenarios, specific categories of images are scarce and difficult to obtain, posing significant challenges to the promotion of supervised learning algorithms [8], [9]. Consequently, the challenge of classifying remote sensing images is further escalating.

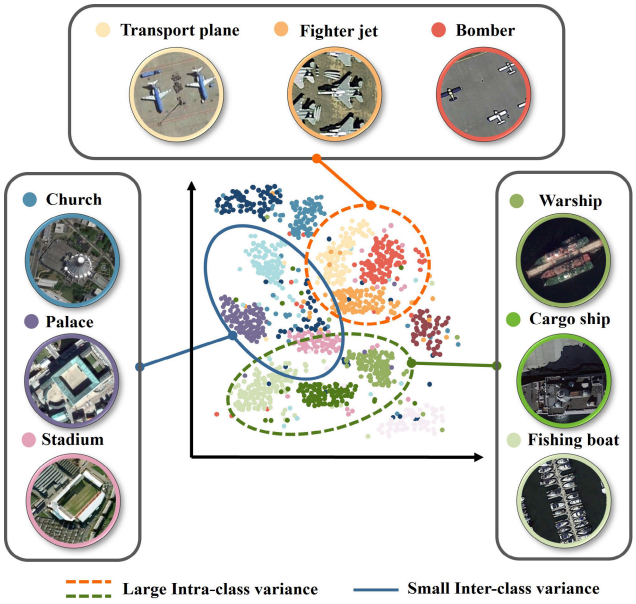


Fig. 1. t-SNE visualization results pertain to selected categories within the NWPU-RESISC45 dataset. In the realm of remote sensing, images within the same class often demonstrate significant diversity, while features among different classes can exhibit remarkable similarities. This inherent variability can introduce ambiguity in the mapping of category features by metric networks, presenting a challenge to accurate classification.

In recent years, deep learning has surfaced as a potent technique for acquiring and depicting image features, delivering significant accomplishments in the field of remote sensing image classification [10], [11], [12], [13], [14]. Within the realm of deep learning methods, convolutional neural networks (CNNs) are prominently recognized as highly favored and successful models. They exhibit exceptional performance in tasks associated with the classification of remote sensing images [15], [16], [17], [18].

In practice, deep learning methods typically necessitate a substantial number of annotated samples for training to ensure that the model acquires robust category generalization capabilities [2], [19]. Insufficient sample quantities frequently

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lead to a lack of diversity in the category features learned by the network, thereby elevating the risk of overfitting. The intricacy of annotating remote sensing images, combined with the presence of considerable background noise and the potential for confusion among closely related category features, magnifies these challenges. Therefore, drawing inspiration from the human ability to rapidly perceive new phenomena, researchers introduced the concept of Few-shot Learning. By utilizing a limited number of labeled samples, the model can be effectively trained, thereby enhancing its ability to generalize to new classes [20], [21], [22], [23], [24], [25].

Among numerous few-shot learning methods, metric learning stands out due to its relatively low computational complexity, strong performance, and minimal resource dependency when faced with new tasks. To be more specific, metric learning aims to establish intricate relationships between data by directly assessing the similarity among samples, thereby facilitating enhanced data classification. Metric learning emphasizes the development of unique distance metrics, which play a crucial role in determining the similarity between support and query images [26]. Illustrative examples of such metrics encompass Euclidean distance, cosine similarity, Markov distance, and Manhattan distance, among others. Nevertheless, despite the potential advantages inherent to metric learning, the direct application of metric learning methods to intricate remote sensing scenarios presents a multitude of challenges [27]. We selected specific classes from the NWPU-RESISC45 dataset [28] and visualized their feature distributions, as depicted in Fig. 1. Observations indicate that the aircraft category, which includes transport planes, bombers, and fighter jets, exhibits significant shape diversity. A similar diversity is observed within the ship category. Furthermore, structures such as stadiums, churches, and palaces often display similar appearances, leading to closely distributed features in the feature space. These visualization results suggest that metric learning methods may struggle to directly utilize distance similarity for effective classification of remote sensing scenes. Hence, purifying the extracted features is a crucial step before classifying images, as it enhances their representativeness for specific categories.

To address the potential challenges mentioned above, we drew inspiration from the principles of metric learning and devised a remote sensing few-shot learning framework named ACL-Net, employing ResNet12 [29] as the backbone network. Our primary emphasis lies in the extraction of representative features rather than the creation of distinctive distance metrics. We utilize the attention-based feature optimization (ABFO) module to aid the model in accurately determining the location of the target region and to mitigate the risk of overfitting to irrelevant features. On one hand, this includes the hybrid self-attention (HSA) module, which applies self-attention globally and locally to improve the model's capacity to capture and integrate features across various scales within an image. On the other hand, it incorporates the information interaction mutual attention (IIMA) module, which employs mutual attention between pairs of images to underscore interimage feature correlation, thus enhancing the model's ability to generalize from a limited number of examples. Moreover, in addressing the challenge of indistinct category features within remote sensing scenes, we applied a dictionary-based contrastive loss (DBCL) module. This module aims to refine the demarcation between categories, aggregate intraclass features, and enhance the discriminative capacity of the features. These improvements enable the metric model to more effectively differentiate between different categories. Finally, the end-to-end network is trained under the proposed training methodologies on five publicly accessible datasets of remote sensing and achieves effective classification of the query set samples. The experimental results validate the effectiveness of our method for Few-shot scene classification in remote sensing scenarios.

The article's contributions can be briefly summarized as follows.

1) We introduce an end-to-end framework for few-shot classification in remote sensing scenes, achieving outstanding performance across five publicly accessible datasets dedicated to few-shot remote sensing scenarios.
2) The ABFO module is introduced during the feature enhancement phase to aid the model in extracting effective features specific to categories from complex remote sensing scenes. It accomplishes this by assigning higher importance to target regions through the HSA module and capturing the correlation between the query set and support set via the IIMA module.
3) We propose a feature-optimized method that incorporates a DBCL module. A category feature vector dictionary is utilized to expand the sparse category vectors within the batch, leading to better training convergence and more reliable results.

The organization of the forthcoming sections in this article is as follows. Section II delves into related work. While Section III offers a comprehensive description of our proposed method. Section IV highlights the results from experiments conducted on various publicly accessible remote sensing datasets. Ultimately, Section V provides the article's concluding remarks.

II. Related Works

In this section, we offer a concise overview of image classification within the context of remote sensing, few-shot learning, and contrastive loss.

A. Remote Sensing Image Scene Classification

Over the past few years, deep learning has garnered remarkable success in computer vision and has steadily made its way into the realm of remote sensing image processing [30], [31], [32]. Deep learning models commonly include CNN, RNN, and autoencoders. These models enable end-to-end training, transforming data into classification results. Of these models, CNNs have emerged as the dominant choice, thanks to their potent feature extraction capabilities [28], [33], [34].

Penatti et al. [35] proposed an approach that leveraged deep features, employing pretrained CNN models for the extraction of features from remote sensing images. Hu et al. [36] combined model adaptation and fine-tuning using AlexNet as the base network, effectively reducing domain differences. Diao et al. [37] introduced deep belief networks (DBNs) to remote sensing image classification, they used a

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saliency mechanism to locate objects quickly and a restricted Boltzmann machine (RBM) to extract features.

Nogueira et al. [38] introduced an incremental approach to remote sensing scene classification by adding new layers or units after pretraining the model, thereby increasing model capacity or complexity. Wang et al. [39] proposed an LSTM network specifically designed for feature extraction from high-resolution images. Liu et al. [40] integrated multitask learning with attention mechanisms, thereby enhancing feature generalization and feature discrimination in the context of remote sensing. Cai and Wei [41] integrated cross-attention mechanisms with graph convolutions to boost both the semantic and spatial information within the features. Shenming et al. [42] proposed the use of conditional random field (CRF) to eliminate noise and refine boundaries, resulting in smoother and more accurate classification results.

Although these aforementioned approaches have gained extensive usage in image classification within remote sensing scenarios and have demonstrated outstanding performance in specific contexts, they rely on sufficient data support to avoid overfitting. In addition, these methods often struggle to adapt and generalize well to newly emerged or sparsely represented classes.

B. Few-Shot Learning

The acquisition and annotation of remote sensing images are typically expensive, time-consuming, and challenging due to the complex and diverse nature of the scenes, where overlapping or ambiguous categories may exist. Therefore, enhancing the classification performance with a restricted amount of labeled remote sensing data has become an urgent problem to solve. In recent years, few-shot learning has gained widespread traction in the remote sensing domain as an effective approach for addressing these challenges [43].

Transfer learning involves pretraining a deep neural network on an extensive dataset and then fine-tuning its parameters on a few-shot dataset to adapt to new categories. This method takes advantage of the advanced feature representation offered by the pretrained model, but it can also suffer from overfitting or catastrophic forgetting due to significant distribution differences between the datasets [44], [45].

The fundamental challenge in few-shot learning is the scarcity of available samples. Data augmentation or feature enhancement on the original data, without altering the network structure, is an efficient approach for addressing the few-shot problem by enriching the diversity of samples. Lin et al. [24] were the first to apply generative adversarial networks (GANs) in the field of remote sensing. They improved DCGAN by incorporating multiple feature layers and introducing a perceptually-aware loss function to enhance the realism of the generated samples. As GANs have evolved, numerous variants have emerged. To achieve a stable generation of high-resolution remote sensing images, Ma et al. [46] progressively increased the complexity of the generator and discriminator modules based on Wasserstein GAN (WGAN). While GANs effectively alleviate distribution differences, increase data volume, and enhance classification performance, the training process of GANs is complex and unstable. It requires careful adjustment of numerous hyperparameters and network structures, which can lead to mode collapse, gradient vanishing, and other issues. Moreover, the quality of generated images by GANs also varies, altering the data distribution and impacting the learning performance of models.

Meta-learning aims to address the few-shot learning problem at a more profound level. Meta-learning strives to facilitate rapid adaptation of the model to novel tasks by learning from multiple different yet related tasks, thus acquiring the ability for autonomous learning [47]. Finn et al. [48] introduced the MAML based on meta-learning, which acquired knowledge from the distribution of all tasks to identify the task-specific parameters within the neural network. By fine-tuning these parameters, the loss function exhibited swift convergence. Liu et al. [49] employed meta-learning to learn a hyperparameter for each layer and developed multiple classifiers based on MAML to address the instability issue of a single classifier.

Meta-learning requires ensuring the relevance between source tasks and target tasks and involves designing appropriate optimization algorithms and loss functions to maintain a balanced performance of the model across various tasks. However, in remote sensing scenarios, there can be significant differences among images from different regions, seasons, and sensors, making it challenging to meet the requirements of meta-learning tasks.

Metric learning refers to training a model to learn an effective and discriminative feature representation or similarity metric from limited samples [50]. The initial metric learning primarily relied on comparisons between samples, where the input comprised multiple samples, and the output represented their similarity. Koch et al. [51] initially proposed the Siamese network for metric learning, utilizing a twin network structure with shared weights to map sample images into a target space. However, the comparison method frequently relies on pairs of samples and necessitates the design of a comparison function. In scenarios with intricate data distributions and expansive datasets, the efficacy of this approach is frequently compromised. Hence, researchers embed sample features into a low-dimensional space to compute their similarity, thereby significantly reducing computational complexity and enabling adaptive learning of embedding functions. Vinyals et al. [52] proposed matching networks, where the model utilized LSTM to map sample information into a low-dimensional vector space and employed kernel density estimation to compute the similarity between new samples and training samples. Snell et al. [53] introduced prototypical networks, which embedded input images into a vector space and utilized the mean of samples from each class as the prototype for that specific class.

Metric learning characterizes the relationships between different image categories through similarity measures without the need for an extensive amount of labeled data or pretrained models, thereby improving model robustness and generalization [54].

However, metric learning relies on the assumption that the prototype features of each class have minimal differences

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in space. Significant variations within class features can negatively impact the performance of metric learning [55], [56].

The previously mentioned methods frequently encounter challenges in fully exploiting the intricate semantic relationships within and among classes, leading to diminished learning efficiency and accuracy in remote sensing scenarios. Our approach offers a more nuanced and potent solution by meticulously analyzing images via the HSA module and fostering richer feature interactions between the support set and query set through the IIMA module. This not only addresses the inherent challenge of limited samples but also maximizes the potential of existing data, thereby notably enhancing classification accuracy and model versatility.

C. Contrastive Learning

Contrastive learning entails comparing samples of the same class as well as pairs of samples from different classes to learn feature representations of the original data, thereby enhancing the performance of downstream tasks. It develops models for measuring similarity, with the aim of maximizing the distinctiveness of category boundaries in the feature embedding space. This approach effectively mitigates the challenge of feature confusion, especially when dealing with challenging negative samples [57], [58]. Common contrastive learning utilizes a contrastive loss function for optimization, which can be expressed as

$$L_{\text{con}} = \frac{1}{2N}\sum_{n=1}^{N}\left[y_n d_n^2 + (1-y_n)\max(\alpha - d_n, 0)^2\right] \quad (1)$$

N represents the number of sample pairs, y_n is a binary label indicating whether the sample pairs belong to the same category, d_n is the distance between sample pairs, and α represents a margin parameter that governs the level of separation between categories.

Given the limited number of samples in few-shot learning, traditional classification algorithms may face the issue of overfitting. Contrastive learning, through the process of comparison, enables the model to allocate increased attention to the feature distinctions among each category, thereby reducing the risk of overfitting and improving the model's generalization ability. Chen et al. [58] introduced the SimCLR, which integrated data augmentation and nonlinear mapping. He et al. [59] proposed MoCo. It utilized a momentum encoder and a queue memory to generate and maintain negative samples, addressing the challenges of negative sample quantity and consistency. In the few-shot scenarios, characterized by a lack of diversity and varying image quality within the batch, the effectiveness of the contrastive loss in terms of generalization often faces challenges.

Traditional contrastive learning methods primarily depend on pairwise comparisons, neglecting to explicitly account for balance or the overall category structure within the batch. Conversely, our approach offers a more comprehensive and balanced framework, leveraging the sample distribution of the entire batch via a category dictionary. This approach enables a deeper and more meticulous understanding of the data, facilitating effective learning of class distribution knowledge from limited samples.

III. Methodology

In this study, taking into account the challenges presented by small-sized and easily occluded targets in remote sensing images, along with the substantial impact of background interference, we introduce an attention-based contrastive learning network (ACL-Net), a metric learning-based network architecture. The framework leverages the ABFO attention module to locally emphasize target object information and employs the DBCL module to maintain clear boundaries between categories, ensuring distinctiveness among them.

A. Overview of the Proposed Method

The comprehensive framework is depicted in Fig. 2. Our proposed framework comprises three key steps: feature extraction, feature enhancement, and category boundary reinforcement. The feature extraction process adopts the ResNet12 structure, consisting of four residual blocks to obtain the initial image feature map $X \in R^{C\times H\times W}$. C, H, and W represent the channel, height, and width dimensions of the residual block feature map before any pooling, respectively. Subsequently, the acquired feature map undergoes improvement during the feature enhancement process named the ABFO module, which includes the hybrid self-attention (HSA) module and the IIMA module. Following this, before inputting the feature vectors into the metric classifier for similarity comparison, we pass them through the DBCL module. This category boundary reinforcement process effectively mitigates metric bias issues within and between classes.

B. Problem Definition

Our work follows the requirements of few-shot classification in remote sensing scenarios, which involve three distinct sets: the training set D_{train}, validation set D_{val}, and test set D_{test}. In contrast to traditional classification approaches, the training, validation, and test sets are mutually exclusive, meaning they do not share any common category. Our specific dataset segmentation is shown in Table I. It means $D_{\text{train}} \cap D_{\text{val}} = \phi$, $D_{\text{val}} \cap D_{\text{test}} = \phi$, $D_{\text{train}} \cap D_{\text{test}} = \phi$.

During the learning phase, the ACL-Net framework is trained on a given dataset D_{train}, which is carefully partitioned into support sets (L_s) and query sets (L_q). This partitioning is designed to simulate the few-shot learning scenario during training: Support set (L_s) consists of $C \times K$ labeled samples, where C is the number of classes, and K represents the number of labeled examples per class. This set serves as the basis for the model to learn the characteristics and variances within each class. Query set (L_q) comprises $C \times Q$ samples, where Q denotes the number of query images per category that the model must classify. The query set necessitates the model to apply the knowledge learned from the support set to accurately classify these new samples belonging to the same category as the support set. After preprocessing the initial image, we obtained the support set and query set to extract the initial image information and support set label information.

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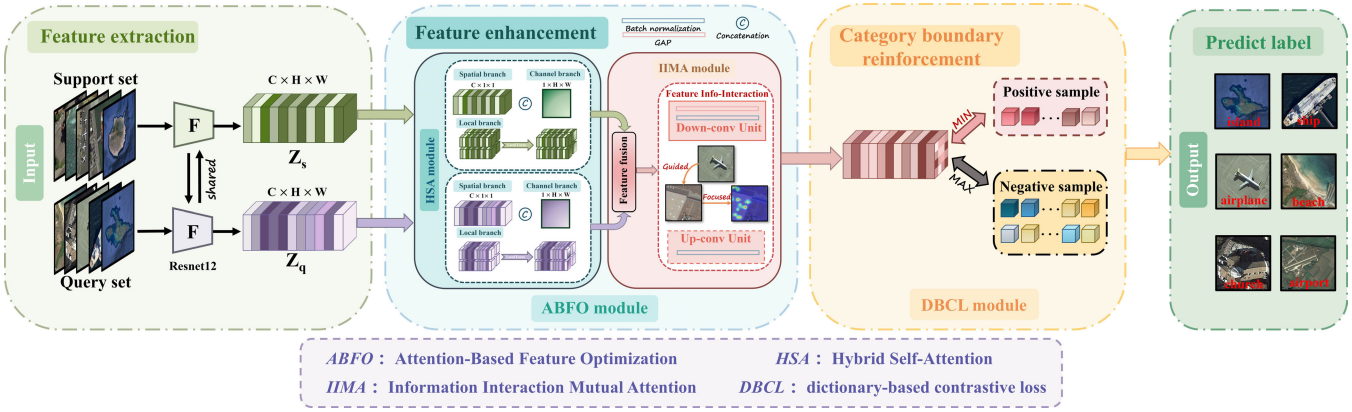


Fig. 2. Overall architecture of ACL-Net. The feature extraction process converts the initial image into feature vectors represented as Z_s and Z_q. The feature enhancement process, facilitated by the ABFO module, comprises the HSA module and the IIMA module. The HSA module captures semantic relationships within features, while the IIMA module fosters interactions among features of the same class, resulting in enhanced feature vectors represented as s and q. The category boundary reinforcement process, executed by the DBCL module, sharpens the boundaries between classes. The final output is then fed into the prototype network for classification.

Subsequently, our model utilizes ResNet12 to extract features, enhances feature representation using the ABFO module, and improves its discriminative ability using the DBCL module. Finally, it embeds the final support set and query set feature vectors into the feature space for measurement and classifies the query set images based on similarity. Through iterative training, the model adeptly extracts features from both the support set and query set, performs similarity measurements, and effectively achieves few-shot image classification tasks in remote sensing scenarios. Similar partitioning is applied to the validation set D_{val} and test set D_{test}. Tasks in the testing and validation phases typically involve assignments similar to those in the training set, albeit with classes that did not appear during the training process. The model adjusts to these novel classes by utilizing the support set, bypassing gradient updates, and thereby directly predicting the labels of the query set examples.

C. ABFO Module

In this work, to achieve the goal of obtaining effective and accurate feature representations using a small amount of data, we introduce a novel ABFO. The ABFO module consists of two components: the HSA module and the IIMA module.

The HSA module calculates weighted feature representations by assessing correlations across the spatial, channel, and local dimensions within feature maps. This comprehensive approach ensures detailed and context-aware feature enhancement, which is crucial for accurate image classification. The IIMA module facilitates the interaction between the support set and the query set, generating a cross-attention map. This enables information exchange and integration between the two sets, further enhancing the representation of features.

Through the designed attention mechanism module, our features are effectively optimized, enhancing both the interpretability and discriminability of the features. This optimization contributes to the improved performance of subsequent network classification tasks.

1) HSA Module: After the feature extractor obtains the initial feature vectors $X \in R^{C\times H\times W}$, we enhance the feature vectors using the HSA Module, and the specific process is illustrated in Fig. 3. The architecture consists of two primary branches: the global attention branch, comprising a channel branch and a spatial branch, and the local attention branch.

a) Global attention branch: The channel branch is responsible for calculating channel attention, which can reflect the semantic hierarchy of the features. The spatial branch, on the other hand, handles the computation of spatial attention, which captures the positional relationships among the features. The design of these two modules is inspired by the concept of polarized filtering [60]. To ensure precise pixel regression, we partition the feature space into polarized components. This approach enhances our ability to capture both long-range dependencies and fine-grained details effectively. In the channel branch, the input feature map is entirely reduced along the spatial dimension, preserving only the channel dimension. Conversely, in the spatial branch, the input feature map is fully compressed along the channel dimension, retaining only the spatial dimension. This approach can effectively maintain high internal resolution while decreasing computational complexity.

The computation process of the channel branch is as follows:

$$A_s^{ch}(\mathrm{X}) = R\left[C_z^{1\times1}\left(\lambda_1\left(C_k^{1\times1}(\mathrm{X})\right) \times SM\left(\lambda_2\left(C_q^{1\times1}(\mathrm{X})\right)\right)\right] \quad (2)$$

where λ_1 and λ_2 are tensor reshape operations, $C_q^{1\times1}$, $C_k^{1\times1}$ and $C_z^{1\times1}$, respectively, perform 1×1 convolution operations. R and SM serve as the Relu and Softmax activation function, ultimately resulting in the calculation of $A_s^{ch} \in R^{C\times1\times1}$.

The final output of the channel branch $Z_s^{ch} = A_s^{ch} \odot X$ is obtained by taking the dot product with the original feature vector X. The formula for calculating the spatial branch is as follows:

$$A_s^{sp}(\mathrm{X}) = R\left[\lambda_1 SM\left(\left(\lambda_2\left(\mathrm{Avg}\left(C_q^{1\times1}(\mathrm{X})\right)\right) \times \lambda_3\left(C_k^{1\times1}(\mathrm{X})\right)\right)\right] \quad (3)$$

where λ_1, λ_2, and λ_3 are tensor reshape operations, $C_q^{1\times1}$ and $C_k^{1\times1}$, respectively, perform 1×1 convolution operations. R and SM serves as the Relu and Softmax activation function,

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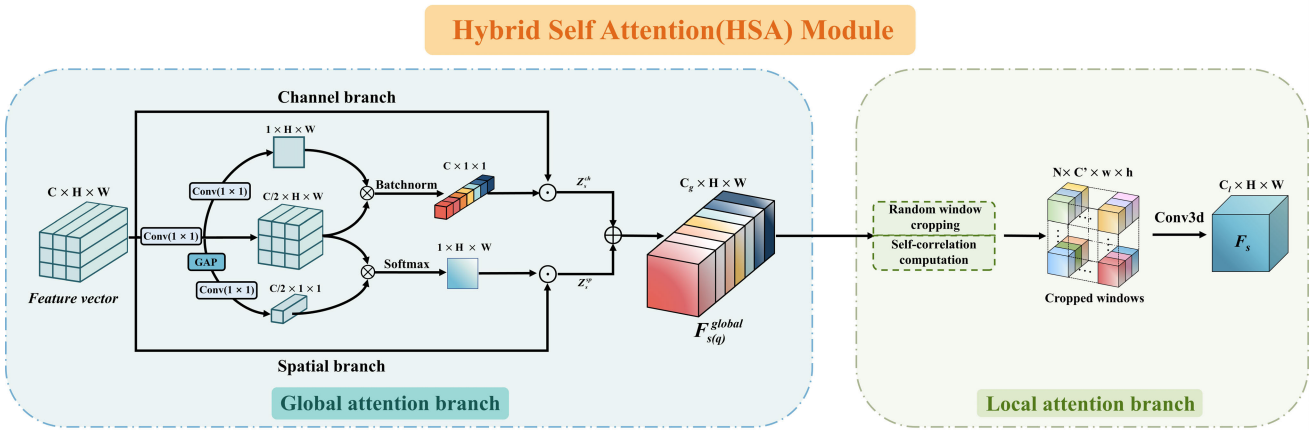


Fig. 3. Structure of HSA module. The module comprises two constituents: a global attention branch, tasked with capturing long-range dependencies, and a local attention branch, which records detailed local information.

Avg signifies the global average pooling (GAP) operation, ultimately resulting in the calculation of $A_s^{sp} \in R^{1\times H\times W}$.

Similar to the channel branch, the final output of the spatial branch $Z_s^{sp} = A_s^{sp} \odot X$ is acquired by performing a dot product operation with the original feature vector X. The aforementioned two branches utilize 1×1 convolutions to compute channel and spatial attention, without employing 3×3 or larger convolution kernels, which offers the advantage of reducing computational cost while preserving high-resolution feature representations.

The outputs of the channel branch and spatial branch $F_s^{\text{global}} \in R^{C_g\times H\times W}$ are summed together to form the output of the global self-attention branch

$$F_s^{\text{global}} = Z_s^{ch} + Z_s^{sp}. \tag{4}$$

b) Local attention branch: The attention mechanism mentioned above belongs to the global self-attention module, which can explore complex global structural information in remote sensing images. However, it may overlook some details and local features. In order to capture the intricate texture features present in remote sensing images, employing a local attention module to interact with crucial information becomes imperative.

The local attention branch in this study is designed to amplify the model's focus on locally significant features within the input image. It dynamically adjusts the emphasis on specific regions of the feature map, enabling precise extraction of local patterns and textures essential for accurate image classification. This mechanism entails partitioning the input feature map into smaller patches, followed by self-correlation analysis to highlight the recurrence of local features. By the window cropping operation, we decompose the feature map into n patches of local specific size $w \times h$. The essence of the local attention mechanism lies in computing self-correlation within each patch using a Hadamard product.

This is accomplished by reshaping the dimensions of each patch and performing element-wise multiplication with the global attention feature map $F_s^{global} \in R^{C_g\times H\times W}$. The original feature map acts as a reference, ensuring that the multiplication emphasizes the intrinsic features of each patch without external biases. The specific calculation process is as follows:

$$F_s = \sum_n \frac{x}{\|x\|} \odot \frac{\text{Reshape}[c(x)]}{\|\text{Reshape}[c(x)]\|} \tag{5}$$

where x denotes the position of elements in the F_s^{global} feature map, Reshape denotes the process of reshaping each patch, $c(*)$ represents the window cropping operation on the feature map, and $\| * \|$ represents the norm operation.

After self-correlation computation, the module uses 1×1 and 3×3 convolutions, with BatchNorm for stability and ReLU for nonlinearity, to capture channel and spatial features. This results in a final representation $F_s \in R^{C_l\times H\times W}$, ready for integration with global features, enhancing local pattern recognition and overall image understanding.

By integrating global attention and local attention, we can enhance the dependencies and contrasts between features, thereby improving feature distinctiveness. This approach enables the simultaneous utilization of channel information, spatial information, and fine-grained texture features, thereby enhancing the richness and expressive power of feature representation. It is particularly beneficial for assisting the model in focusing on target features and improving classification performance.

2) IIMA Module: The IIMA module is designed to interact between the query set and support set information, following the HSA module. The module generates a cross-attention map, denoted as $A_s(A_q)$, which serves to emphasize the regions corresponding to the target objects. This approach can enhance the distinctiveness of the extracted features, thereby assisting the neural network in learning image features of the same category with greater effectiveness. Fig. 4 provides an illustration of the structure of our proposed IIMA module.

The diagram is primarily composed of two key modules: the Feature map fusion module and the cross-attention computation module.

a) Feature map fusion module: After obtaining the enhanced support set feature map $F_s \in R^{C_l\times H_s\times W_s}$ and the query set feature map $F_q \in R^{C_l\times H_q\times W_q}$ from the HSA module, the feature map undergoes standardization across the channel dimension followed by a 1×1 convolution operation. This transforms the channel dimension C_l into a more

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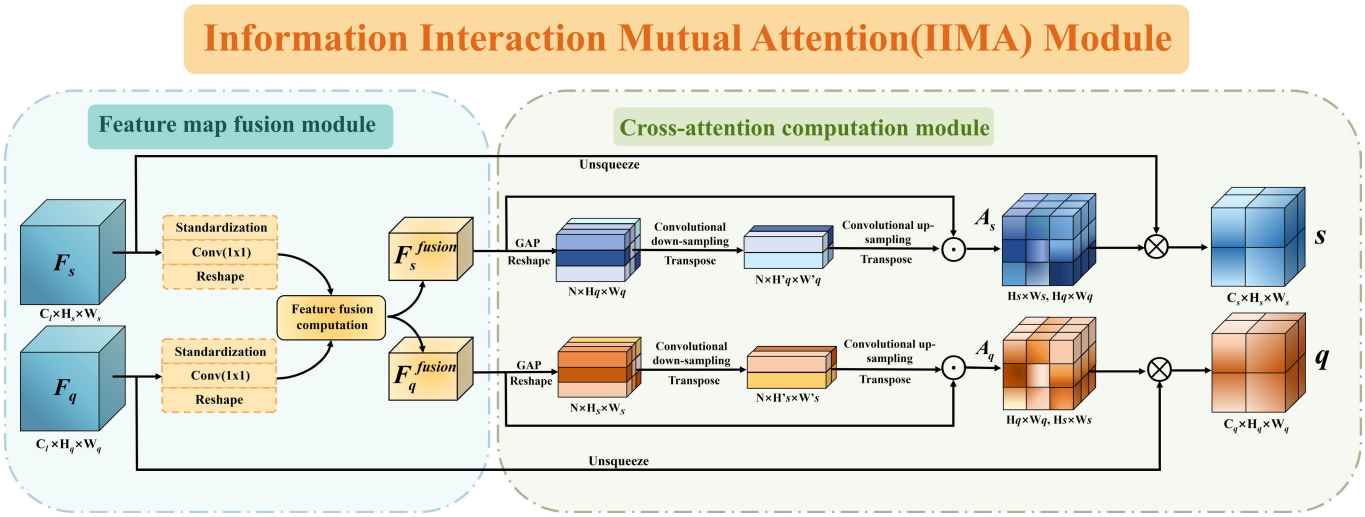


Fig. 4. Structure of IIMA module. It integrates the enhanced features and facilitates interactions between the target features of the support set and the query set, generating the cross-attention map $A_s(A_q)$. The matrix multiplication operation between this map and the previously obtained self-attention map leads to the final optimized attention map, represented as $s(q)$.

compact representation C_l' and computes the correlation matrix between them, resulting in a four-dimensional correlation tensor $F_{fusion} \in R^{H_s \times W_s \times H_q \times W_q}$

$$F_{\text{fusion}}(x_s, x_q) = \text{sim}(F_s, F_q) \tag{6}$$

where x_s and x_q denote the spatial positions within the feature map, while sim is representative of the cosine similarity function. To facilitate readers' clear and intuitive comprehension of the feature map processing within the IIMA module, we denote the original correlation tensor F_{fusion} as F_s^{fusion} and F_q^{fusion} in Fig. 4. It's important to note that these representations are identical.

b) Cross-attention computation module: This module can identify target regions within the fusion feature map where the support set and query set share similarities, thereby enhancing the features of these target regions. This allows the network model to pay more attention to these informative regions. Take the support set as an example. In this attention module, the feature map F_s^{fusion} undergoes a GAP operation, which eliminates the H_s and W_s dimensions of the feature map. Subsequently, the resulting tensor is passed through two separate convolution modules, one for downsampling and the other for upsampling. The formula for updating the weights of the target regions is as follows:

$$\omega = \text{conv}(\text{Relu}(\text{GAP}(F_s^{\text{fusion}}))) \tag{7}$$

where ω denotes the weights associated with the target region, while Relu signifies a nonlinear activation function.

The final attention map $A_s \in R^{H_s \times W_s \times H_q \times W_q}$ is generated by multiplying the feature map F_s^{fusion} with the weight-enhanced attention map F_{att} and applying the Softmax activation function. The calculation process for A_s is as follows:

$$A_s(x_s) = \frac{1}{HW} \sum_{x_q} \frac{\exp\left(C\left(w^T x_s, w^T x_q\right)/\tau\right)}{\sum_{x_s}^{H_s \times W_s} \exp\left(C\left(w^T x_s, w^T x_q\right)/\tau\right)} \tag{8}$$

where x_s and x_q are the positions in the final feature map, τ refers to the temperature factor, $C(w^T x_q, w^T x_s)$ represents the matching score obtained by calculating the similarity between the weighted values of x_s and x_q.

The final embedding feature map in the support set denoted as $s \in R^{C_s \times H \times W}$, is derived through an element-wise multiplication operation applied to the attention map A_s, which is generated by the IIMA module, and the original feature map $F_s \in R^{C_l \times H_s \times W_s}$, which is produced by the HSA module

$$s = \sum_{x_s} A_s(x_s) F_s(x_s). \tag{9}$$

The embedding feature map $q \in R^{C_q \times H \times W}$ in the query set is also obtained in the same way, and it can be represented as

$$q = \sum_{x_q} A_q\left(x_q\right) F_q\left(x_q\right). \tag{10}$$

In the IIMA module, information interaction is employed to generate a cross-attention map. Specifically, within the target region, the module amalgamates image information derived from both the query set and the support set. This process underscores the significance or relevance of features within the support and query sets for that specific region.

D. Dictionary-Based Contrastive Loss

In this article, after the ABFO module, a contrastive loss function based on a dictionary is incorporated during network training. This approach combines the concept of a prototype network and utilizes label information to select contrast sample pairs. The goal is to cluster categories that share the same semantic information in the embedding space, meanwhile, ensuring that categories with different semantic information are distinctly separated in the space.

In few-shot learning scenarios, the limited diversity of samples within a batch frequently leads to an insufficient representation of the overall characteristics of each category

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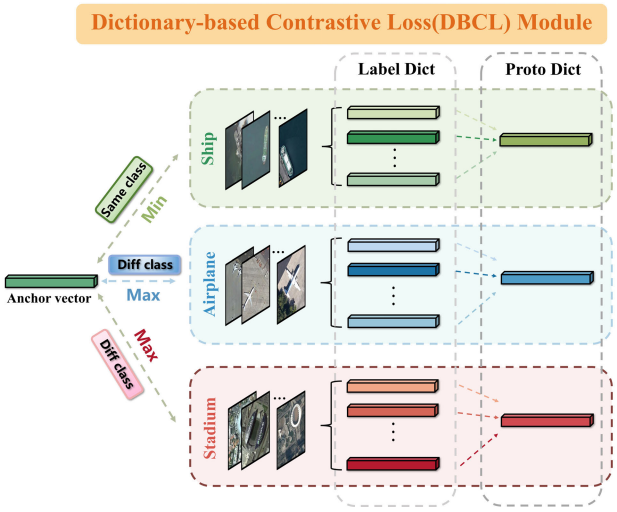


Fig. 5. Calculation process for the DBCL.

of samples. Therefore, if the computation of contrastive loss solely takes into account the relationships between sample pairs within a batch, while neglecting the overarching structure of the feature space, it could potentially result in suboptimal solutions.

Therefore, in order to expand the number of sample pairs and mitigate the risk of loss calculation divergence, we devised a dictionary with category labels as keys, and the corresponding values are sets of feature vectors for each label. As the batch continuously updates, the label dictionary dynamically expands. During each iteration, the anchor vector is no longer juxtaposed with other vectors within the batch. Instead, it is systematically contrasted with vectors originating from both identical and distinct classes retrieved from the dictionary.

Meanwhile, to ensure the training process's robustness, we aggregate and average the feature vectors corresponding to each category in the category dictionary, forming prototypes for each category. These prototypes are stored in the prototype dictionary and participate in the calculation of comparative losses. By capturing the basic features of each category, prototypes effectively mitigate the influence of values exhibiting significant personal biases during model training. However, as the category vector dictionary rapidly expands with each iteration of the batch, it can significantly affect the code's running speed. Therefore, we've set an upper limit of the number of category vectors in the dictionary at 1000. When the count of category vectors hits this limit, we replace the initial vector in the category dictionary with the current vector.

Throughout the training process, every sample within a batch serves as an anchor point. Employing the label information, we query the two aforementioned dictionaries to retrieve sets of both positive and negative samples that share the same label. Our final DBCL function is expressed as follows:

$$L_{\mathrm{c}} = -\frac{1}{N}\sum_{N}\log\left[\frac{\exp\left(\mathrm{sim}\left(i_a, i_{\mathrm{p}}\right)/\alpha\right)}{\sum_{k\neq i}\exp(\mathrm{sim}(i_a, i_{\mathrm{n}})/\alpha) + \exp\left(\mathrm{sim}\left(i_a, i_{\mathrm{p}}\right)/\alpha\right)}\right] \tag{11}$$

where i_a denotes the set of anchor samples present within a batch, N represents the total number of samples in the batch, α is the temperature coefficient, $\mathrm{sim}(i_a, i_p)$ represents the similarity between positive sample pairs, and $\mathrm{sim}(i_a, i_n)$ signifies the similarity between negative sample pairs

$$\mathrm{sim}(i_a, i_p) = \mathrm{sim}(i_a, {i_{\mathrm{pro}}}^{+}) + \omega\left[\frac{\sum_{\mathrm{dict}}^{n-1}\mathrm{sim}(i_a, i_{\mathrm{dict}}^{+})}{n-1}\right] \tag{12}$$

where n denotes the total number of samples present within the label dictionary, $\mathrm{sim}(i_a, {i_{\mathrm{pro}}}^{+})$ is the similarity between an anchor point and the prototype of the same class in prototype dictionary, $\mathrm{sim}(i_a, i_{\mathrm{dict}}^{+})$ is the similarity between an anchor point and a vector from the same class in the label dictionary, and ω represent the weight of prototype dictionary similarity

$$\mathrm{sim}(i_a, i_n) = \mathrm{sim}(i_a, {i_{\mathrm{pro}}}^{-}) + \omega\left[\frac{\sum_{\mathrm{dict}}^{n-1}\mathrm{sim}(i_a, i_{\mathrm{dict}}^{-})}{n-1}\right] \tag{13}$$

where $\mathrm{sim}(i_a, {i_{\mathrm{pro}}}^{-})$ is the similarity between an anchor point and the prototype of a different class in the prototype dictionary, $\mathrm{sim}(i_a, i_{\mathrm{dict}}^{-})$ be the similarity between an anchor point and a vector from a different class in the dictionary. The detailed calculation steps are illustrated in Fig. 5.

Ultimately, the network training is steered by combining the contrastive loss function, denoted as L_c, with the cross-entropy loss, represented by L_{cross} [61]. This combination not only guides the training process but also performs gradient updates to optimize the model's performance. The cross-entropy loss specifically classifies the anchor points themselves. The procedure for calculating the cross-entropy loss is as follows:

$$L_{\mathrm{cross}} = -\log\frac{\exp(\omega_i^T x_q + b_i)}{\sum_{\hat{i}}^{D_{\mathrm{train}}}\exp(\omega_{\hat{i}}^T x_q + b_{\hat{i}})} \tag{14}$$

where ω and b represent the weights and biases, respectively.

The ultimate loss function can be expressed as follows:

$$L = L_{\mathrm{cross}} + \sigma L_c \tag{15}$$

where σ is a hyperparameter that balances the weight of the contrastive loss. During the testing and inference phase, only the L_{cross} loss is calculated, and the parameters of the fully connected layer remain frozen.

Experimental evidence has demonstrated that our contrastive learning method can acquire more discriminative and generalizable feature representations, thereby enhancing the model's capability to handle complex image scenarios.

IV. Experiments

To demonstrate the effectiveness of our ACL-Net, we conducted experiments on five publicly available remote sensing datasets: NWPU-RESISC45 [28], WHU-RS19 [62], UC-Merced Land Use datasets [63], AID dataset [64], and EuroSAT dataset [65].

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TABLE I
STANDARD SEGMENTATION REPRESENTATION OF REMOTE SENSING SCENE CLASSIFICATION DATASET

Dataset	Training set	Validation set	Test set
NWPU-RESISC45	Church; Mobile home park; Stadium; Roundabout; Chaparral; Airplane; Cloud; Ship; Golf course; Meadow; Harbor; Freeway; Lake; Wetland Baseball diamond; Island; Railway; Mountain; Sparse residential; Palace; Bridge; Desert Sea ice; Beach; Rectangular farmland	Thermal power station; Storage tank; Terrace; Railway station; Tennis court; Snowberg; Industrial area; Runway; Overpass; Commercial area	River; Parking lot; Ground track field; Medium residential; Dense residential; Basketball court; Circular farmland Intersection; Airport; Forest
WHU-RS19	Parking; Port; Residential; Bridge; Industrial; Mountain; Airport; Football field; Desert;	Railway station; Beach; Forest; Farmland; Park	Viaduct; Pond; Meadow; River; Commercial
UCM	Parking lot; Dense residential; Harbor; Chaparral; Buildings; Overpass; Agricultural; Medium residential; Baseball diamond; Freeway	Intersection; Storage tanks; Airplane; Runway; Forest;	Golf course; River; Sparse residential; Tennis court; Beach; Mobile home park;
AID	Bare Land; Baseball Field; Beach; Bridge; Commercial; Dense Residential; Farmland; Meadow; Pond; Port; Resort; Stadium Airport; Desert; Railway Station	Medium Residential; Mountain; Park; Parking; Playground	River; Viaduct Square; Storage Tanks; School; Square; Forest; Industrial; Center; Church; Sparse Residential
EuroSAT	Annual Crop; Forest; Herbaceous Vegetation; Highway; Industrial;	Pasture; Permanent Crop;	Residential; river; Sea Lake;

A. Dataset Introduction

The experimental setup and protocols for the first three datasets adhered to the standard segmentation and evaluation rules established by DLA-MatchNet [66]. The partitioning rules for the AID dataset align with those defined by MKN [67]. However, in the absence of set partitioning rules for the EuroSAT dataset, we conducted the division ourselves. The specific details of this division are illustrated in Table I.

The *NWPU-RESISC45* dataset serves as a publicly accessible dataset for the remote sensing scene classification of images. Comprising 31 500 RGB images at a 256×256 pixel resolution. It encompasses a total of 45 scene categories, each containing 700 images. For standard training, we partitioned the dataset into 25 categories, allocating ten categories for validation and an additional ten categories for testing.

The *WHU-RS19* dataset encompasses remote sensing images belonging to 19 distinct categories. The dataset contains a total of 1005 RGB images with a resolution of 256×256 pixels, with 50 images per category. Specifically, our allocation strategy involved using nine categories for training, five categories for validation, and an additional five categories for testing.

The *UC-Merced* dataset is tailored for land use remote sensing scene recognition. Encompassing 21 distinct scene categories, this dataset comprises 100 images within each category, accumulating to a total of 2100 images. Our approach to utilizing the dataset involved partitioning it into ten categories for training while reserving six categories for testing and an additional five categories for validation purposes.

The *AID* dataset is a comprehensive collection of aerial images obtained from Google Earth. Comprising 10 000 images distributed across 30 categories such as airports, bare land, baseball fields, beaches, and others, it offers a diverse range of scenes. The dataset incorporates multisource images, sourced from various remote sensing imaging sensors. Our approach to dataset utilization included partitioning it into 15 categories for training, reserving ten categories for testing, and allocating an additional five categories for validation purposes.

The *EuroSAT* dataset is a benchmark for land use and land cover classification. It's based on Sentinel satellite images, covering 13 spectral bands and consisting of ten classes with a total of 27 000 labeled and geo-referenced images. The EuroSAT dataset has been used in various Earth observation applications, demonstrating its value in improving geographical maps. We partitioned the dataset into five categories for training, two categories for validation, and allocated the remaining three categories for testing.

B. Parameter Setting

In our experiments, we utilized the PyTorch framework and trained the models using a 3080Ti GPU. Before input into the backbone network, the original image is uniformly resized to a fixed size of 84×84 pixels. For feature extraction, we employed ResNet12, which consists of four basic residual blocks.

To enhance the contrastive loss module, we employed data augmentation techniques, including random cropping and horizontal mirroring. The final chosen hyperparameter settings are outlined below: the temperature coefficient τ for the IIMA module was set to 6, the temperature coefficient α for the DBCL module was set to 1.5, and the loss weight hyperparameter σ was set to 0.05. Furthermore, we utilized the SGD optimizer to refine the parameters of the network model, setting a learning rate of 0.08 and a weight decay parameter of $5e-4$.

We conducted few-shot classification experiments in two scenarios: five-way five-shot and five-way one-shot. During the testing phase, we performed 2000 random samplings on the test set and averaged the results for inference.

C. Experimental Results and Comparisons

To clearly illustrate the advantages of our ACL-Net, we conducted a direct comparison of the final classifica-

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TABLE II

CLASSIFICATION ACCURACY, MODEL PARAMETERS, AND FPS VALUES FOR DIFFERENT MODELS IN THE FIVE-WAY ONE-SHOT SCENARIO ACROSS FIVE DATASETS. **BOLD** AND UNDERLINE DENOTE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Year	5-way 1-shot						
		NWPU-RESISC45	WHU-RS19	UCM	AID	EuroSAT	Params	FPS
MatchingNet [52]	NeurIPS-2016	40.31±0.13	51.25±0.61	34.68±0.91	42.17±0.78	51.63±0.32	10.72M	19.81
ProtoNet [53]	NeurIPS-2017	41.38±0.26	58.17±0.56	52.34±0.19	49.91±0.47	58.69±0.24	11.19M	25.23
MAML [48]	ICML-2017	48.04±0.21	50.87±0.23	48.94±0.31	48.16±0.37	60.13±0.27	9.21M	18.34
Meta-SGD [74]	arXiv-2017	40.96±0.08	51.78±1.05	51.13±0.95	50.01±0.93	51.98±0.55	18.97M	16.78
RelationNet [68]	CVPR-2018	66.21±0.28	61.74±0.51	48.48±0.75	53.51±0.68	73.82±0.48	27.07M	23.34
DLA-MatchNet [66]	TGRS-2021	68.80±0.70	68.27±1.83	53.76±0.62	53.41±0.99	-	50.91M	31.36
SCL-MLNet [69]	TGRS-2022	62.21±1.12	63.36±0.88	51.37±0.79	59.46±0.96	-	191.59M	33.17
TSC [70]	ISPRS2022	73.26±0.15	70.99±0.74	55.11±0.68	68.21±0.83	-	-	34.36
MKN [67]	TGRS-2022	65.84±0.89	-	58.45±0.54	57.29±0.59	-	-	-
MPCL-Net [71]	TGRS-2023	55.94±0.04	61.84±0.12	56.46±0.21	60.61±0.43	81.27±0.36	45.01M	19.80
TDNET [72]	TGRS-2023	65.85±0.53	64.24±0.51	-	67.48±0.51	-	**8.33M**	26.02
TeAW [73]	ICASSP-2023	70.23±0.45	-	56.94±0.39	70.35±0.44	-	-	-
ACL-Net	**Ours**	**76.13±0.24**	**78.30±0.32**	**59.74±0.46**	**70.86±0.31**	**88.75±0.36**	**13.94M**	**39.65**

TABLE III

CLASSIFICATION ACCURACY, MODEL PARAMETERS, AND FPS VALUES FOR DIFFERENT MODELS IN THE FIVE-WAY FIVE-SHOT SCENARIO ACROSS FIVE DATASETS (THREE-WAY FOR EUROSAT). **BOLD** AND UNDERLINE DENOTE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Year	5-way 5-shot (3-way for EuroSAT)						
		NWPU-RESISC45	WHU-RS19	UCM	AID	EuroSAT	Params	FPS
MatchingNet [52]	NeurIPS-2016	47.27±0.28	54.36±0.38	53.34±0.17	52.34±0.89	57.24±0.43	10.72M	19.81
ProtoNet [53]	NeurIPS-2017	62.77±0.14	80.54±0.42	69.28±0.67	70.48±0.21	77.54±0.46	11.19M	25.23
MAML [48]	ICML-2017	62.98±0.47	64.26±0.32	60.61±0.29	63.13±0.94	71.48±0.31	9.21M	18.34
Meta-SGD [74]	arXiv-2017	47.46±0.37	65.47±0.65	63.68±0.59	65.31±0.38	56.82±0.37	18.97M	16.78
RelationNet [68]	CVPR-2018	78.37±0.28	79.15±0.35	62.17±0.33	68.65±0.95	85.74±0.57	27.07M	23.34
DLA-MatchNet [66]	TGRS-2021	81.63±0.46	79.89±0.33	63.01±0.51	70.61±0.36	-	50.91M	31.36
SCL-MLNet [69]	TGRS-2022	80.86±0.76	77.62±0.81	68.09±0.92	76.31±0.68	-	191.59M	33.17
TSC [70]	ISPRS2022	84.62±0.35	82.18±0.32	69.20±0.64	81.72±0.54	-	-	34.36
MKN [67]	TGRS-2022	82.67±0.55	-	**77.92±0.23**	75.42±0.31	-	-	-
MPCL-Net [71]	TGRS-2023	76.24±0.12	80.34±0.54	76.57±0.07	76.78±0.08	89.12±0.26	45.01M	19.80
TDNET [72]	TGRS-2023	82.16±0.32	84.15±0.32	-	80.56±0.36	-	**8.33M**	26.02
TeAW [73]	ICASSP-2023	85.57±0.25	-	77.50±0.27	84.62±0.25	-	-	-
ACL-Net	**Ours**	**86.54±0.23**	**90.43±0.15**	74.89±0.29	**85.37±0.34**	**95.46±0.15**	13.94M	**39.65**

tion accuracy, model parameters, and frames per second (FPS) values during the testing inference stage with other few-shot classification methods, including SCL-MLNet [69], DLA-MatchNet [66], Meta-SGD [74], MatchingNet [52], MAML [48], Relation Network [68], Prototypical Network [53], TSC [70], MKN [67], MPCL-Net [71], TDNet [72], and TeAW [73]. All methods were assessed under uniform experimental settings. Prototypical network [53], MatchingNet [52], relation network [68], MKN [67], and TDNet [72], akin to our approach, are rooted in the concept of metric learning. DLA-MatchNet [66] incorporates attention mechanisms, while SCL-MLNet [69] and TSC [70] are grounded in contrastive learning principles. TeAW [73] and MPCL-Net [71] are founded on multimodal learning. Tables II and III present the comparisons between our method and the aforementioned approaches on the five datasets, considering scenarios of five-way five-shot and five-way one-shot.

Table II presents the results for the five-way one-shot scenario, and it is evident that in this scenario, our ACL-Net model achieved the highest accuracy across all five datasets compared to other models. On the NWPU and WHU datasets, our model achieved improvements of 2.87% and 7.31%, respectively, compared to other models with the highest accuracy.

As indicated in Table III, in the five-way five-shot scenario, our ACL-Net exhibited the highest accuracy across all datasets, except UCM. Notably, the WHU-RS19 dataset showcased a remarkable 6.28% accuracy improvement compared to the models with the highest accuracy, underscoring the superior classification performance of our model on this dataset. Therefore, the positive outcomes indicate that our approach exhibits greater competitiveness in the one-shot scenario as opposed to the five-shot scenario. Moreover, the model displays a more distinct advantage in addressing situations with extremely limited samples.

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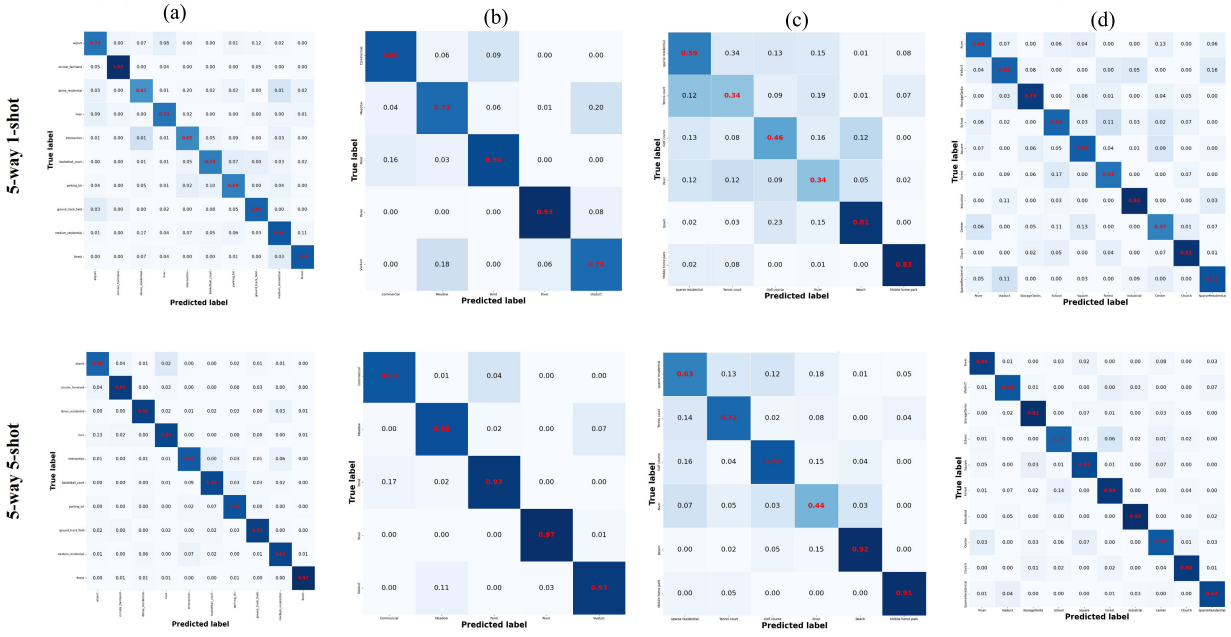


Fig. 6. Visualization of the confusion matrices on the four experimental datasets. (a) NWPU-RESISC45. (b) WHU-RS19. (c) UCM. (d) AID.

TABLE IV
ABLATION EXPERIMENTS ON THE HYPERPARAMETER σ OF THE DBCL LOSS MODULE ON THE NWPU-RESISC45 DATASET

σ	0.01	0.02	0.03	0.04	**0.05**	0.06	0.07	0.08	0.09	0.1
5-way 1-shot	71.62±0.38	72.24±0.45	73.32±0.19	74.92±0.33	**76.13±0.24**	74.18±0.41	73.96±0.42	73.25±0.12	72.36±0.25	70.78±0.32
5-way 5-shot	83.37±0.32	83.89±0.28	84.67±0.21	85.75±0.19	**86.54±0.23**	86.08±0.36	85.21±0.24	84.85±0.27	84.06±0.29	83.55±0.32

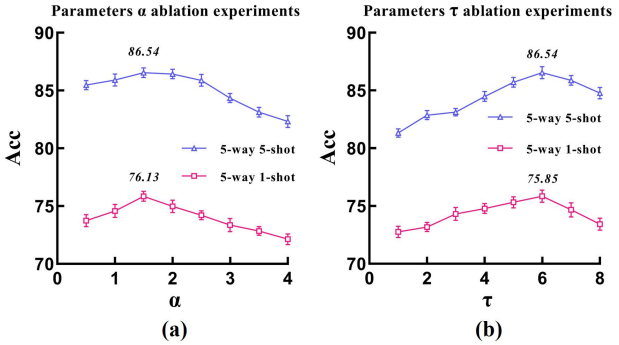


Fig. 7. Temperature coefficient ablation experiments on the NWPU-RESISC45 dataset. (a) Temperature coefficient α for the DBCL module. (b) Temperature coefficient τ for the IIMA module.

In addition, we conducted a comparison of FPS values and the number of parameters across all accessible models. It is evident that, in comparison to other models, we attained an FPS value nearing 40 while upholding the model's complexity, ensuring a satisfactory execution time for the algorithm.

Furthermore, for a visual representation of our classification accuracy for each category, we conducted a comparison between the predicted categories generated by our method and the true categories across the four datasets, excluding the EuroSAT dataset. This comparison resulted in confusion matrices. The visualization results of matrices for the two experimental scenarios are presented in Fig. 6. For instance, on the NWPU-RESISC45 dataset, the accuracy for the forest class attained 97% in the five-way five-shot scenario, indicating an approximate 10% enhancement compared to the TSC [70] model.

In summary, the proposed ACL-Net in this study demonstrates exceptional performance across five publicly accessible remote sensing scene datasets specifically designed for few-shot learning. The ABFO attention mechanism in our model is employed to concurrently enhance the features of target objects and emphasize their positions. This capability enables the network to extract more discriminative features. Moreover, for the refinement of class prototypes within the prototype network, we introduced a contrastive loss module based on the label vector dictionary. This module reduces the distance between intraclass vectors while widening the gap between interclass vectors, thereby significantly enhancing the model's precise classification capability for query set samples.

D. Hyperparameter Ablation Experiment

To guarantee the synergy of the three hyperparameters set in this study for achieving the highest accuracy, we conducted ablation experiments specifically on the values of each hyperparameter using the NWPU-RESISC45 dataset.

To determine a suitable weight allocation for our DBCL contrastive loss relative to the cross-entropy loss, Table IV presents the results of our ablation experiment, where the weight σ is assigned to the DBCL loss. It is evident that when

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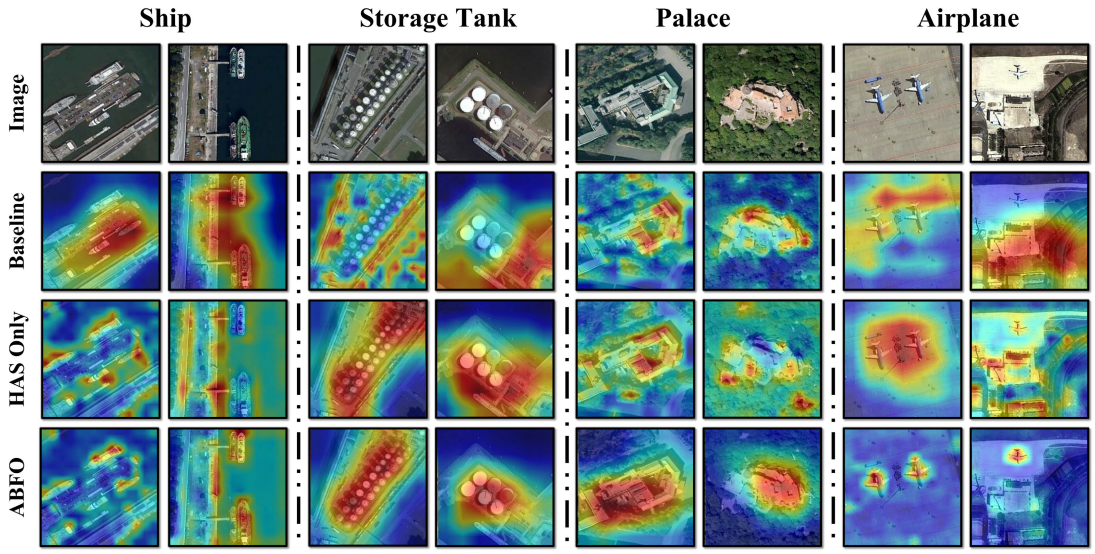


Fig. 8. Visualization of CAM for four classic remote sensing scenes. Each class is presented through four distinct sections, displaying the original image, the CAM image following feature extraction by ResNet12, the CAM image after feature enhancement via the HSA module, and finally, the CAM image postfeature optimization using the whole attention module (ABFO).

TABLE V
ABLATION EXPERIMENTS CONDUCTED ON THE NWPU-RESISC45 DATASET

#	HSA	IIMA	DBCL	1-shot	5-shot
(a)	✗	✗	✗	68.46± 0.38	80.57± 0.53
(b)	✓	✗	✗	69.51± 0.26	81.81± 0.56
(c)	✗	✓	✗	73.43± 0.21	83.97± 0.45
(d)	✗	✗	✓	71.35± 0.84	82.47± 0.64
(e)	✓	✓	✗	75.34± 0.36	85.35± 0.42
(f)	✓	✗	✓	73.81± 0.44	84.17± 0.27
(g)	✓	✓	✓	**76.13±0.24**	**86.41 ± 0.49**

TABLE VI
ABLATION EXPERIMENTS CONDUCTED ON THE WHU-RS19 DATASET

#	HSA	IIMA	DBCL	1-shot	5-shot
(a)	✗	✗	✗	62.67± 0.73	74.78± 0.49
(b)	✓	✗	✗	65.38± 0.27	79.09± 0.55
(c)	✗	✓	✗	75.24± 0.35	86.33± 0.84
(d)	✗	✗	✓	70.68± 0.34	81.13± 0.37
(e)	✓	✓	✗	77.12± 0.83	88.87± 0.40
(f)	✓	✗	✓	72.75± 0.36	86.41± 0.14
(g)	✓	✓	✓	**78.30±0.32**	**90.43±0.15**

the hyperparameter σ is set to 0.05, the model attains optimal performance. Any deviation from this parameter, either increasing or decreasing, results in a decline in performance.

Moreover, ablation experiments were conducted on the temperature coefficients of both the IIMA module and the DBCL module to deduce the optimal values. The curves depicting the final prediction accuracy with the variation of the two temperature coefficients are illustrated in Fig. 7. The results depicted in the figure indicate that the trends remain consistent in both five-way five-shot and five-way one-shot scenarios, with the accuracy peaking simultaneously at $\alpha = 1.5$, $\tau = 6$. The presented experimental results were obtained while maintaining the other two hyperparameters at their optimal values, thus exploring the optimal performance boundary for the third parameter.

TABLE VII
ABLATION EXPERIMENTS CONDUCTED ON THE UCM DATASET

#	HSA	IIMA	DBCL	1-shot	5-shot
(a)	✗	✗	✗	55.63± 0.85	68.05± 0.59
(b)	✓	✗	✗	57.79± 0.83	69.21± 0.45
(c)	✗	✓	✗	57.71± 0.74	72.21± 0.61
(d)	✗	✗	✓	56.37± 0.63	70.85± 0.29
(e)	✓	✓	✗	57.98± 0.68	73.54± 0.60
(f)	✓	✗	✓	57.47± 0.53	72.46± 0.43
(g)	✓	✓	✓	**59.74±0.46**	**74.89±0.29**

E. Module Ablation Experiment

To demonstrate the effectiveness of our ABFO attention module and the DBCL module, we conducted ablation experiments on all modules of our ACL-Net on the NWPU-RESISC45, WHU-RS19, and UCM datasets. These experiments were performed to analyze the impact of each module on the classification performance within the framework.

1) Baseline: Removing the ABFO module and the DBCL module from our approach, we retained only the ResNet12 feature extractor and the classifier.

2) Baseline+HSA: The HSA attention module has been incorporated into the baseline to enhance the representation of scene features.

3) Baseline+IIMA: The IIMA attention module has been integrated into the baseline to enable information interaction among samples of the same class.

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TABLE VIII

COMPARING THE CLASSIFICATION ACCURACY OF DISTINCT ATTENTION MECHANISMS ACROSS THE PREVIOUSLY MENTIONED THREE DATASETS WITHIN THE CONTEXT OF A FIVE-WAY ONE-SHOT SCENARIO

Method	Self-Att	Mutual-Att	NWPU-RESISC45	WHU-RS19	UCM
Resnet12 (Baseline)	✗	✗	68.46± 0.38	62.67 ± 0.73	55.63 ± 0.85
SE [75]	✓	✗	68.79 ± 0.21	63.77 ± 0.84	55.87 ± 0.92
SCE [76]	✓	✗	68.92 ± 0.28	63.89 ± 0.38	55.94 ± 0.87
CAN [77]	✗	✓	69.71 ± 0.46	65.73 ± 0.62	57.47 ± 0.43
No-local [78]	✓	✗	68.97 ± 0.50	64.16 ± 0.68	57.34 ± 0.97
Local [79]	✓	✗	68.89 ± 0.71	64.25 ± 0.74	57.25 ± 0.78
CSAM [80]	✓	✗	69.23 ± 0.15	64.01 ± 0.82	56.98 ± 0.54
CBAM [81]	✗	✓	68.87 ± 0.37	64.95 ± 0.58	56.06 ± 0.46
HSA(ours)	✓	✗	69.51 ± 0.26	65.38 ± 0.27	**57.79 ± 0.83**
IIMA(ours)	✗	✓	**73.43 ± 0.21**	**75.24 ± 0.35**	57.71 ± 0.74

TABLE IX

COMPARING THE CLASSIFICATION ACCURACY OF DISTINCT ATTENTION MECHANISMS ACROSS THE PREVIOUSLY MENTIONED THREE DATASETS WITHIN THE CONTEXT OF A FIVE-WAY FIVE-SHOT SCENARIO

Method	Self-Att	Mutual-Att	NWPU-RESISC45	WHU-RS19	UCM
Resnet12 (Baseline)	✗	✗	80.57 ± 0.53	74.78 ± 0.49	68.05 ± 0.59
SE [75]	✓	✗	80.98 ± 0.14	78.46 ± 0.81	68.25 ± 0.61
SCE [76]	✓	✗	81.14 ± 0.74	78.39 ± 0.69	68.71 ± 0.12
CAN [77]	✗	✓	82.64 ± 0.24	80.67 ± 0.33	69.82 ± 0.76
No-local [78]	✓	✗	81.16 ± 0.47	79.88 ± 0.31	69.27 ± 0.24
Local [79]	✓	✗	81.09 ± 0.91	79.93 ± 0.67	69.06 ± 0.75
CSAM [80]	✓	✗	82.34 ± 0.63	79.74 ± 0.63	69.29 ± 0.64
CBAM [81]	✗	✓	80.51 ± 0.28	80.57 ± 0.31	68.01 ± 0.86
HSA(ours)	✓	✗	81.81 ± 0.56	79.09 ± 0.55	69.21 ± 0.45
IIMA(ours)	✗	✓	**83.97 ± 0.45**	**86.33 ± 0.84**	**72.21 ± 0.61**

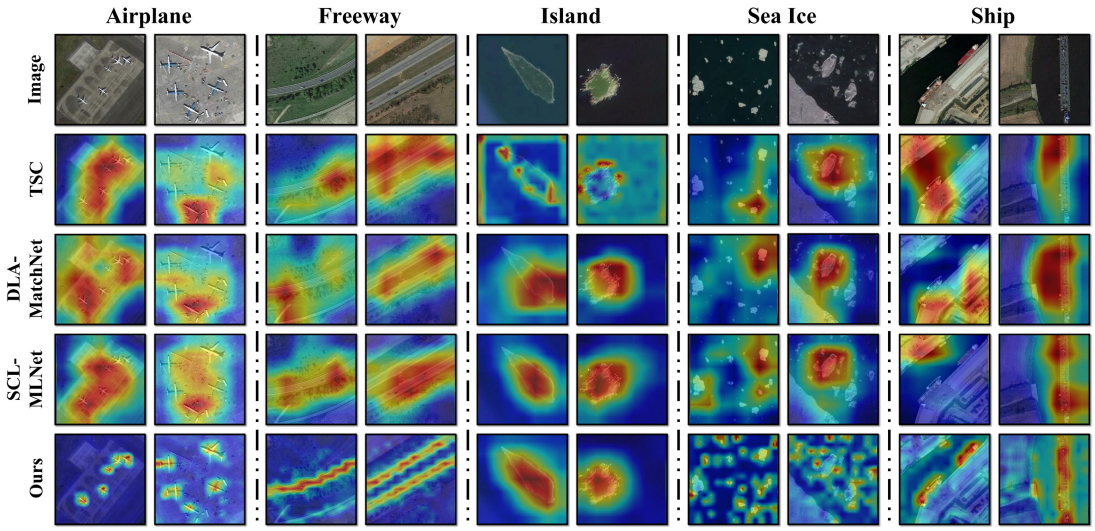


Fig. 9. Comparison of visualization results for the four methods was conducted under a five-way one-shot one-query task. The figure unmistakably illustrates that, in contrast to alternative attention modules, our attention module orchestrates the network's attention toward the finer details of the target objects in a more comprehensive manner, leading to the extraction of more effective features.

4) Baseline+DBCL: The DBCL module has been integrated into the baseline to improve classification performance.

5) Baseline+ABFO: The complete ABFO attention module has been incorporated into the baseline to capture a greater number of discriminative features.

6) Baseline+HSA+DBCL: The HSA and the DBCL module are incorporated into the baseline for global feature enhancement as well as category boundary reinforcement.

7) ACL-Net: The overall network architecture proposed by our work.

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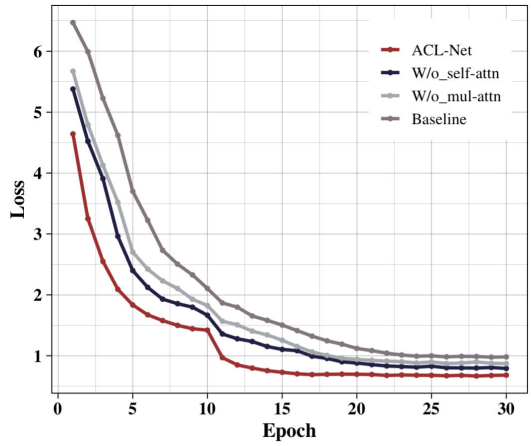


Fig. 10. Variation of training loss under the five-way five-shot scenario, on the NWPU-RESISC45 dataset. The inclusion of the attention mechanism accelerates the convergence speed of the loss and reduces the lower bound of the loss.

As shown in Tables V–VII, our three proposed modules contribute to improving the classification accuracy to varying degrees in the aforementioned three datasets. The baseline+IIMA module exhibits superior performance when compared to the baseline+HSA module. This implies that the IIMA module facilitates efficient information exchange between the support set and query set, leading to a more effective reinforcement of features. In conclusion, the ablation experiment results demonstrate the efficacy of our ACL-Net framework.

To showcase the superiority of our ABFO attention module compared to other classical attention modules, we conducted a comparative analysis of the training results across the aforementioned three datasets mentioned, as shown in Tables VIII–IX. To ensure a fair comparison, all attention modules were tested on the same framework as described in this article, with only the attention module being modified while keeping other parameters constant. The results reveal that our HSA and IIMA attention modules consistently attain the highest classification accuracy in both two scenarios, additionally confirming the advantages and efficacy of our attention module. It's worth noting that in the five-way one-shot scenario with the UCM dataset, HSA demonstrates superior performance when compared to IIMA.

During the testing with the CBAM attention module [81], we observed that our final accuracy surpasses that of DLA-MatchNet, which also employs this module. This indirectly demonstrates the contribution of our ABFO module to the accuracy.

F. Visualization Result

To provide additional evidence of the ABFO attention module's effectiveness, we conducted class activation mapping (CAM) on the feature maps generated by each attention module. The CAM results illustrate the extent of feature optimization achieved by the module. In the CAM images, deeper red indicates higher relevance to the target class, while deeper blue represents lower relevance. We randomly selected four categories: ship, storage tank, palace, and airport, for CAM visualization analysis. The results are shown in Fig. 8.

Based on the visualizations presented in the figures, it is evident that the network, after the ABFO attention module, can effectively focus on the objects themselves, highlighting the target regions. The HSA module assigns weights to the feature vectors, capturing the approximate positions of the targets more accurately. However, relying solely on self-attention can sometimes introduce interference from background factors that are similar to the targets. For example, in the ship and airplane scenes, the self-attention module captures some irrelevant information along with the target information, resulting in incorrect identification of those regions as the "target." The IIMA module addresses this issue by facilitating information exchange among the targets, leading to more discriminative results. To emphasize the strengths of our attention module, under the same scene settings, we performed CAM on the same images and compared the results with TSC, DLA-MatchNet, and SCL-MLNet. The comparison results are presented in Fig. 9.

Upon observation, it becomes apparent that our method, in comparison to alternative approaches, places greater emphasis on capturing intricate details within the target area and provides a more accurate description of the object's contour information. Moreover, it exhibits a higher degree of resilience against background interference. Using the airplane category as an illustrative example, our approach proficiently detects all the airplanes in the image on the right. In contrast, other methods tend to overlook some objects to varying degrees. Across all five categories, our approach excels in delivering intricate object contour descriptions while minimizing attention to extraneous background elements. In summary, our method yields more accurate feature representations when confronted with challenging samples compared to the other three methods.

To demonstrate the impact of the attention module on the classification performance during the training process of our algorithm, we visualized the training loss throughout the process using the NWPU-RESISC45 dataset. The specific results are depicted in Fig. 10. Observations reveal that both our proposed HSA module and IIMA module significantly expedite the convergence of the loss curve and reduce the lower bound of the loss in comparison to the baseline alone. Moreover, the synergistic integration of these two modules results in even more substantial improvements.

To offer a more intuitive illustration of the feature optimization capabilities of our ACL-Net framework. We conducted t-SNE visualization on the class distribution of the WHU dataset under the five-way five-shot 15-query scenario. The particularized results are depicted in Fig. 11. Visualization results affirm that our network adeptly enhances class discrimination. It achieves this by bringing similar features into closer proximity while simultaneously magnifying distinctions between disparate classes.

V. Failure Scenario Analysis

Extensive experiments in Sections IV-C and IV-F demonstrate that our method achieves competitive performance.

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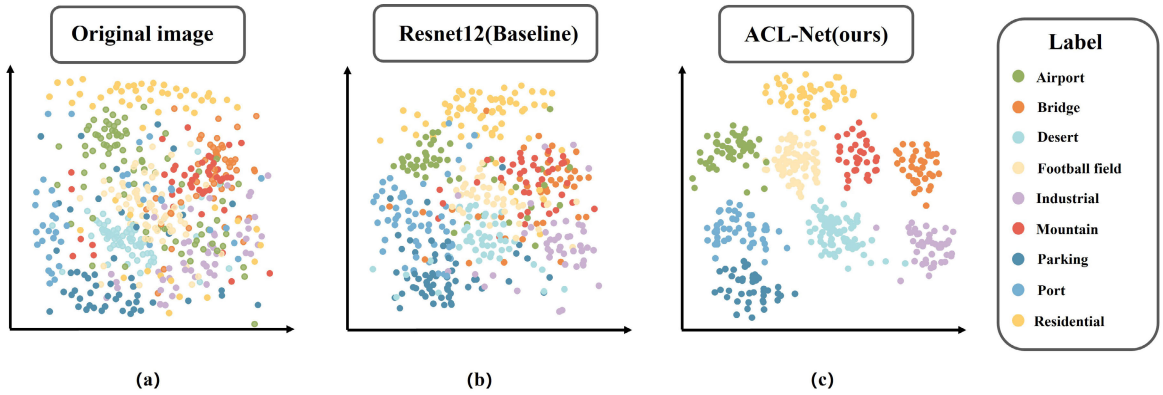


Fig. 11. t-SNE visualization of feature distribution. (a) Feature distribution of original images. (b) Feature distribution extracted by the baseline. (c) Feature distribution extracted by ACL-Net.

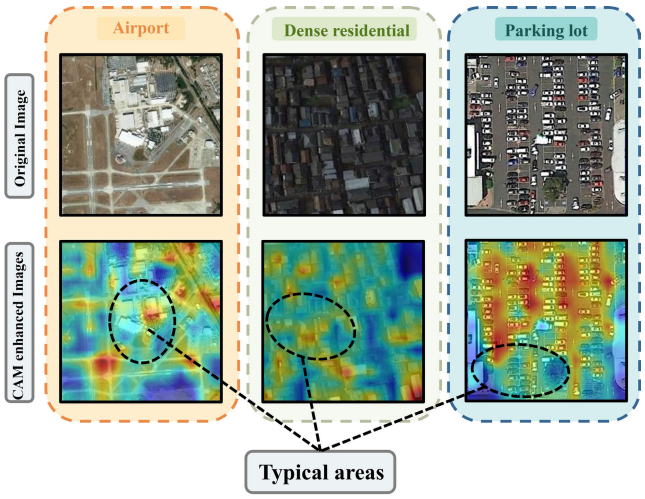


Fig. 12. Visual comparison of complex categories. The target features within the marked typical areas have not received the attention they deserve.

However, there are still some failure cases that need further discussion. Taking the NWPU-RESSC45 dataset as an example, the confusion matrix visualization reveals that the classification accuracy of categories such as airport, dense residential, and parking lot is notably low. As shown in Fig. 12, the visualization results indicate that, due to high category diversity, substantial intraclass complexity, and considerable background interference, our ACL-Net struggles to focus on the target objects themselves in these categories. This limitation leads to a significant deviation in accuracy from the average for these three categories. Future efforts will concentrate on exploring methods to enhance the feature extraction of complex remote sensing image categories.

VI. Conclusion

Within this article, we present an innovative framework for few-shot remote sensing image classification, which we refer to as ACL-Net. This framework comprises three key components: a feature extractor, a feature optimizer, and a classifier. The feature extractor leverages Resnet12 for feature extraction, while the feature optimizer integrates an ABFO attention module to enrich the model's feature representation, emphasizing the target region. Within the classifier component, we employ a DBCL loss module to fine-tune the network's classifier, resulting in well-defined boundaries between categories and, consequently, achieving classification in remote sensing images. We conducted comparative experiments with multiple models on five publicly accessible datasets, and, in most cases, our method achieved SOTA performance. The experimental results further affirm the superiority and feasibility of our approach in the field of few-shot classification in remote sensing.

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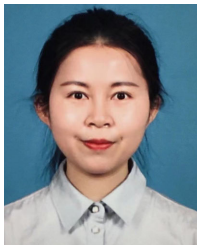
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